114-2 AtmSci 7118 颱風動力 Typhoon Dynamics （郭鴻基、陳柏孚）	
(1) 作業 3 次 (30%)	作業，約二至四週後交。
(2) 期中、期末書面報告 (30%)	期中書面報告（5/13）、期末書面報告（6/14）各一份。 請針對學生報告之任一主題，撰寫書面小論文，內容可為文獻整理、延伸討論與分析、參加會議心得、課堂討論筆記等。 以中文撰寫，至多 3000 字，至多附圖 8 張。 撰寫主題須由非本人報告之議題中選出。
(3) 主題報告（20%）	同學互評，每人有各自的分數： 組織與架構 / 簡報設計 /表達 / 回答與討論 25% / 25% / 25% / 25%
(4) 課堂表現 （20%）	課堂參與度、出席表現，由老師評分。

114-2 AtmSci 7118 颱風動力 Typhoon Dynamics （郭鴻基、陳柏孚）	
Week 1, 2/25	**Lecture 1 (Kuo)**
	▪ Course overview ▪ [Lecture #1] Basic dynamics.
Week 2, 3/4	**Lecture 2 (Kuo)**
	▪ [Lecture #2] Axisymmetric vortex theory fundamentals. ▪ **Homework I**
Week 3, 3/11	**Lecture 3 (Chen)**
	▪ [Lecture #3] How do tropical cyclones intensify? ▪ Confirm report order by lottery
Week 4, 3/18	**Lecture 4 (Chen)**
	▪ [Lecture #4] Tropical cyclone formation
Week 5, 3/25	**Lecture 5 (Chen)**
	▪ [Lecture #5] Tropical cyclone structure and rainbands ▪ Confirm selected papers
Week 6, 4/1	**Lecture 6 (Kuo)**
	▪ [Lecture #6] Tropical cyclone concentric eyewall ▪ **Homework II**
Week 7, 4/8	**Student presentation & Lecture 7**
	▪ Student presentation #1 王裕仁

	▪ [Lecture #7]
Week 8, 4/15	**Student presentation & Lecture 8**
	▪ Student presentation #2 陳楷薪 ▪ [Lecture #8]
Week 9, 4/22	**ICMCS-XVII**
	No class, mid-term project due (4/26 23:59 UTC+8)
Week 10, 4/29	**Student presentation & Lecture 9**
	▪ Student presentation #3 簡孟優 ▪ [Lecture #9] TBD (Chen)
Week 11, 5/6	**Student presentation & LLAT.ty hands-on**
	▪ Student presentation #4 李恆安 ▪ **Homework III** ▪ LLAT.ty hands-on (Chen)
Week 12, 5/13	**Student presentation & Lecture 10**
	▪ Student presentation #5 陳柏翔 ▪ [Lecture #10] TBD (Kuo)
Week 13, 5/20	**Student presentation & Lecture 11**
	▪ Student presentation #6 林芃翰 ▪ [Lecture #11] TBD (Kuo)
Week 14, 5/27	**Student presentation & Lecture 12**
	▪ Student presentation #7 李俊霖 ▪ [Lecture #12]
Week 15, 6/3	**Student presentation & Lecture 13**
	▪ Student presentation #8 葉品辰 ▪ [Lecture #13]
Week 16, 6/10	Final exam week
	No class, final project due (6/14 23:59 UTC+8)

Student presentation topics (suggested)
Topic #1: TC motion
▪ Ito, K., Wu, C. C., Chan, K. T., Toumi, R., & Davis, C. (2020). Recent progress in the fundamental understanding of tropical cyclone motion. *Journal of the Meteorological Society of Japan. Ser. II*, *98*(1), 5-17. ▪ Shi, W., Fei, J., Huang, X., Liu, Y., Ma, Z., & Yang, L. (2016). Rossby wave energy dispersion from tropical cyclone in zonal basic flows. *Journal of Geophysical Research: Atmospheres*, *121*(7), 3120-3138. ▪ Hsu, L. H., Kuo, H. C., & Fovell, R. G. (2013). On the geographic asymmetry of typhoon translation speed across the mountainous island of Taiwan. *Journal of the Atmospheric Sciences*, *70*(4), 1006-1022.
Topic #2: Rapid intensification
▪ Fudeyasu, H., Ito, K., & Miyamoto, Y. (2018). Characteristics of tropical cyclone rapid intensification over the western North Pacific. *Journal of Climate*, *31*(21), 8917-8930. ▪ Xu, J., Wang, Y., & Yang, C. (2019). Factors affecting the variability of maximum potential intensity (MPI) of tropical cyclones over the North Atlantic. *Journal of Geophysical Research: Atmospheres*, *124*(13), 6654-6668. ▪ Tsujino, S., and H. Kuo, 2020: Potential Vorticity Mixing and Rapid Intensification in the Numerically Simulated Supertyphoon Haiyan (2013). *J. Atmos. Sci.*, 77, 2067–2090. ▪ Judt, F., Rios-Berrios, R., & Bryan, G. H. (2023). Marathon versus sprint: Two modes of tropical cyclone rapid intensification in a global convection-permitting simulation. *Monthly Weather Review*, *151*(10), 2683-2699.
Topic #3: PV mixing and intensification
▪ Kuo, H. C., Cheng, W. Y., Yang, Y. T., Hendricks, E. A., & Peng, M. S. (2016). Deep convection in elliptical and polygonal eyewalls of tropical cyclones. *Journal of Geophysical Research: Atmospheres*, *121*(24), 14-456. ▪ Cha, T. Y., Bell, M. M., Lee, W. C., & DesRosiers, A. J. (2020). Polygonal eyewall asymmetries during the rapid intensification of Hurricane Michael (2018). *Geophysical Research Letters*, *47*(15), e2020GL087919. ▪ Cha, T. Y., & Bell, M. M. (2024). Tropical cyclone asymmetric eyewall evolution and intensification in a two-layer model. *Weather and Climate Dynamics*, *5*(3), 1013-1029.
Topic #4: Concentric eyewall in TCs
Topic #5: TC formation
▪ Raymond, D. J., Sessions, S. L., & López Carrillo, C. (2011). Thermodynamics of tropical cyclogenesis in the northwest Pacific. *Journal of Geophysical Research: Atmospheres*, *116*(D18). ▪ Gjorgjievska, S., & Raymond, D. J. (2014). Interaction between dynamics and thermodynamics during tropical cyclogenesis. *Atmospheric Chemistry and Physics*, *14*(6), 3065-3082. ▪ Raymond, D. J., & Kilroy, G. (2019). Control of convection in high-resolution simulations of tropical cyclogenesis. *Journal of Advances in Modeling Earth Systems*, *11*(6), 1582-1599.

Topic #6: Radiation and diurnal cycle in TCs
▪ - ▪
Topic #7: TC size
▪ Wang, D., Lin, Y., & Chavas, D. R. (2022). Tropical cyclone potential size. *Journal of the Atmospheric Sciences*, *79*(11), 3001-3025. ▪ Lu, K. Y., & Chavas, D. R. (2022). Tropical cyclone size is strongly limited by the Rhines scale: Experiments with a barotropic model. *Journal of the Atmospheric Sciences*, *79*(8), 2109-2124. ▪ Zhang, W., Chan, K. T., & Xu, L. (2025). Global Climatology of Rapid Expansion of Tropical Cyclones. *International Journal of Climatology*, *45*(1), e8692.
Topic #8: TC-VWS interaction
▪ Rios-Berrios, R., Finocchio, P. M., Alland, J. J., Chen, X., Fischer, M. S., Stevenson, S. N., & Tao, D. (2024). A review of the interactions between tropical cyclones and environmental vertical wind shear. *Journal of the Atmospheric Sciences*, *81*(4), 713-741.
Topic #9: VWS, LMF and TC structure
▪ Chen, B. F., Davis, C. A., & Kuo, Y. H. (2018). Effects of low-level flow orientation and vertical shear on the structure and intensity of tropical cyclones. *Monthly Weather Review*, *146*(8), 2447-2467. ▪ Chen, B. F., Davis, C. A., & Kuo, Y. H. (2019). An idealized numerical study of shear-relative low-level mean flow on tropical cyclone intensity and size. *Journal of the Atmospheric Sciences*, *76*(8), 2309-2334. ▪ Chen, B. F., Davis, C. A., & Kuo, Y. H. (2021). Examination of the combined effect of deep-layer vertical shear direction and lower-tropospheric mean flow on tropical cyclone intensity and size based on the ERA5 reanalysis. *Monthly Weather Review*, *149*(12), 4057-4076.
Topic #10: Vortex tilting and 3-D vorticity structure
▪ Yu, C. L., Tang, B., & Fovell, R. G. (2023). Tropical cyclone tilt and precession in moderate shear: Precession hiatus in a critical shear regime. *Journal of the Atmospheric Sciences*, *80*(3), 909-932. ▪ Yu, C. L., Tang, B., & Fovell, R. G. (2023). Diverging behaviors of simulated tropical cyclones in moderate vertical wind shear. *Journal of the Atmospheric Sciences*, *80*(12), 2837-2860. ▪ Yu, C. L., Tang, B., & Fovell, R. G. (2024). Tropical Cyclone Boundary Layer Asymmetries in a Tilt-Following Perspective. *Journal of the Atmospheric Sciences*, *81*(9), 1543-1563.
Topic #11: TC climate change
Topic #12: TC prediction